

Exercises 1.8

1. Prove that $n^2 + 1 \geq 2^n$ where n is a positive integer with $1 \leq n \leq 4$.

Solution: $P(n): n^2 + 1 \geq 2^n$

$$P(0): \text{LHS} = 0^2 + 1 = 1 \quad \text{RHS} = 2^0 = 1$$

$\therefore \text{LHS} \geq \text{RHS}$. $P(0)$ is True

$$P(1): \text{LHS} = 1^2 + 1 = 2 \quad \text{RHS} = 2^1 = 2$$

$\therefore \text{LHS} \geq \text{RHS}$. $P(1)$ is True

$$P(2): \text{LHS} = 2^2 + 1 = 5 \quad \text{RHS} = 2^2 = 4$$

$\therefore \text{LHS} \geq \text{RHS}$. $P(2)$ is True

$$P(3): \text{LHS} = 3^2 + 1 = 10 \quad \text{RHS} = 2^3 = 8$$

$\therefore \text{LHS} \geq \text{RHS}$. $P(3)$ is True

$$P(4): \text{LHS} = 4^2 + 1 = 17 \quad \text{RHS} = 2^4 = 16$$

$\therefore \text{LHS} \geq \text{RHS}$. $P(4)$ is True.

$\therefore n^2 + 1 \geq 2^n$, where $1 \leq n \leq 4$, n is an integer. $\square$

2. Use a proof by cases to show that 10 is not the square of a positive integer.

Solution:

The set of positive integers x can either be: 1) $x \leq 4$ or 2) $x \geq 4$.

Case 1:

$$x \leq 4. \quad 1^2 = 1 \neq 10 \quad 2^2 = 4 \neq 10 \quad 3^2 = 9 \neq 10$$

Case 2:

$$x \geq 4. \quad 4^2 = 16 > 10. \text{ It is easy to see that for all } x \geq 4, x^2 > 10.$$

$\therefore 10$ is not the square of a positive integer $\square$

3. Use a proof by cases to show that 100 is not the cube of a positive integer.